Introduction to Embedded

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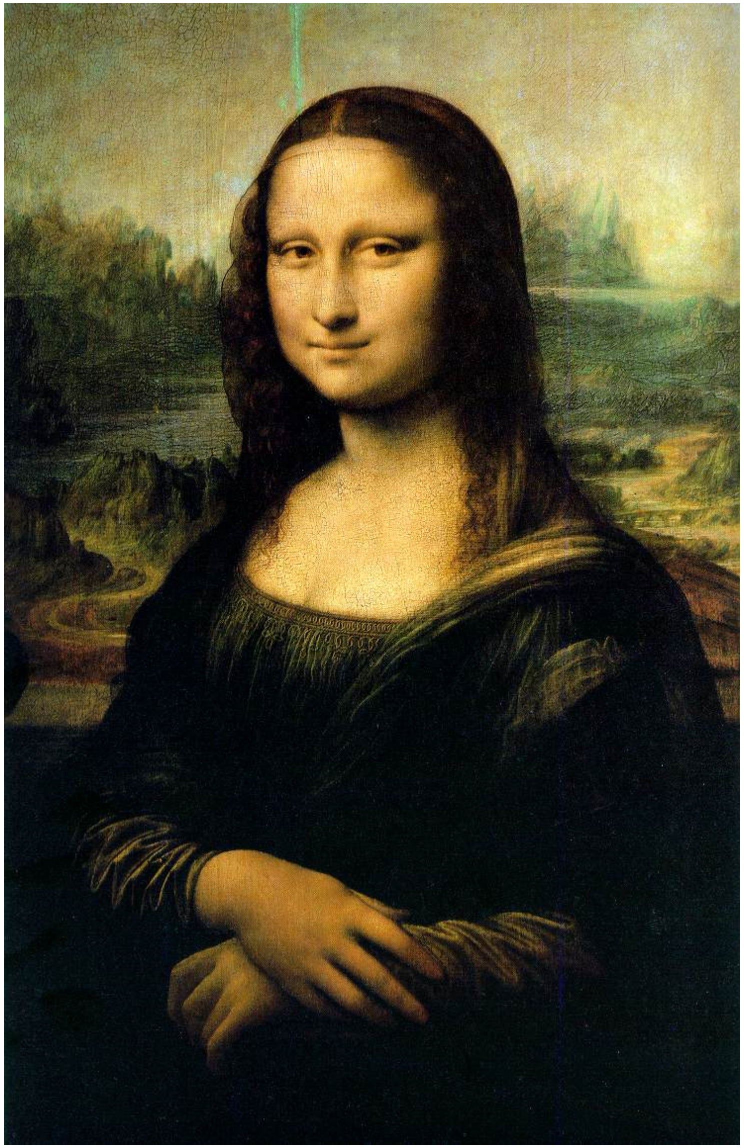
Embedded Solutions

1506

- Leonardo da Vinci (Mona Lisa)



1606

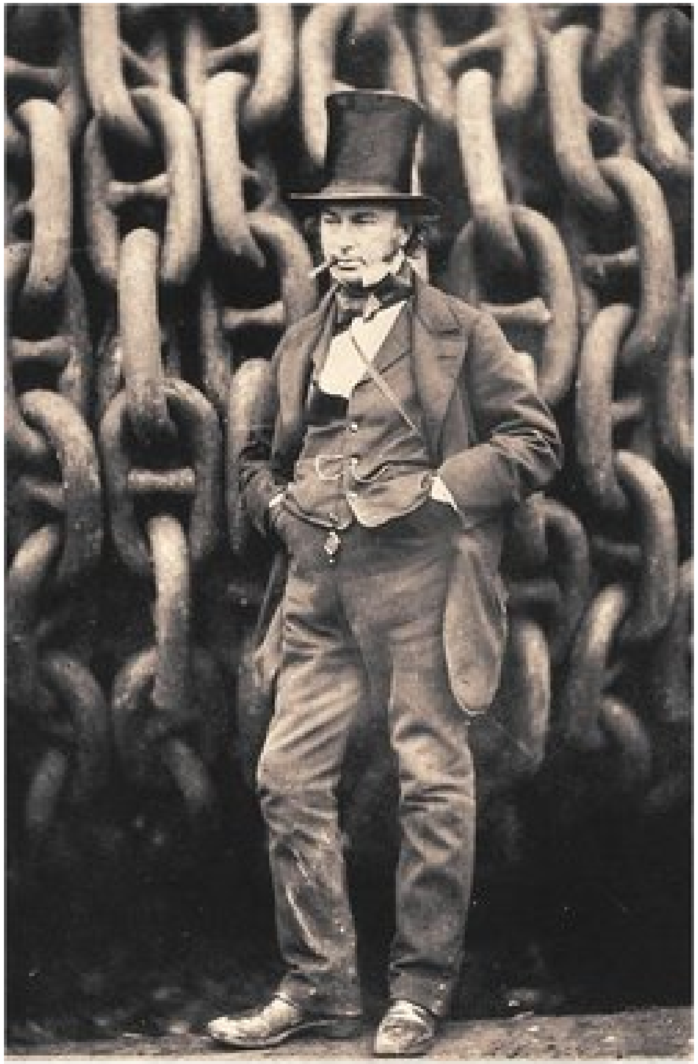
- Gunpowder Plot trial

1706

- Benjamin Franklin born

1806

- Isambard Kingdom Brunel born

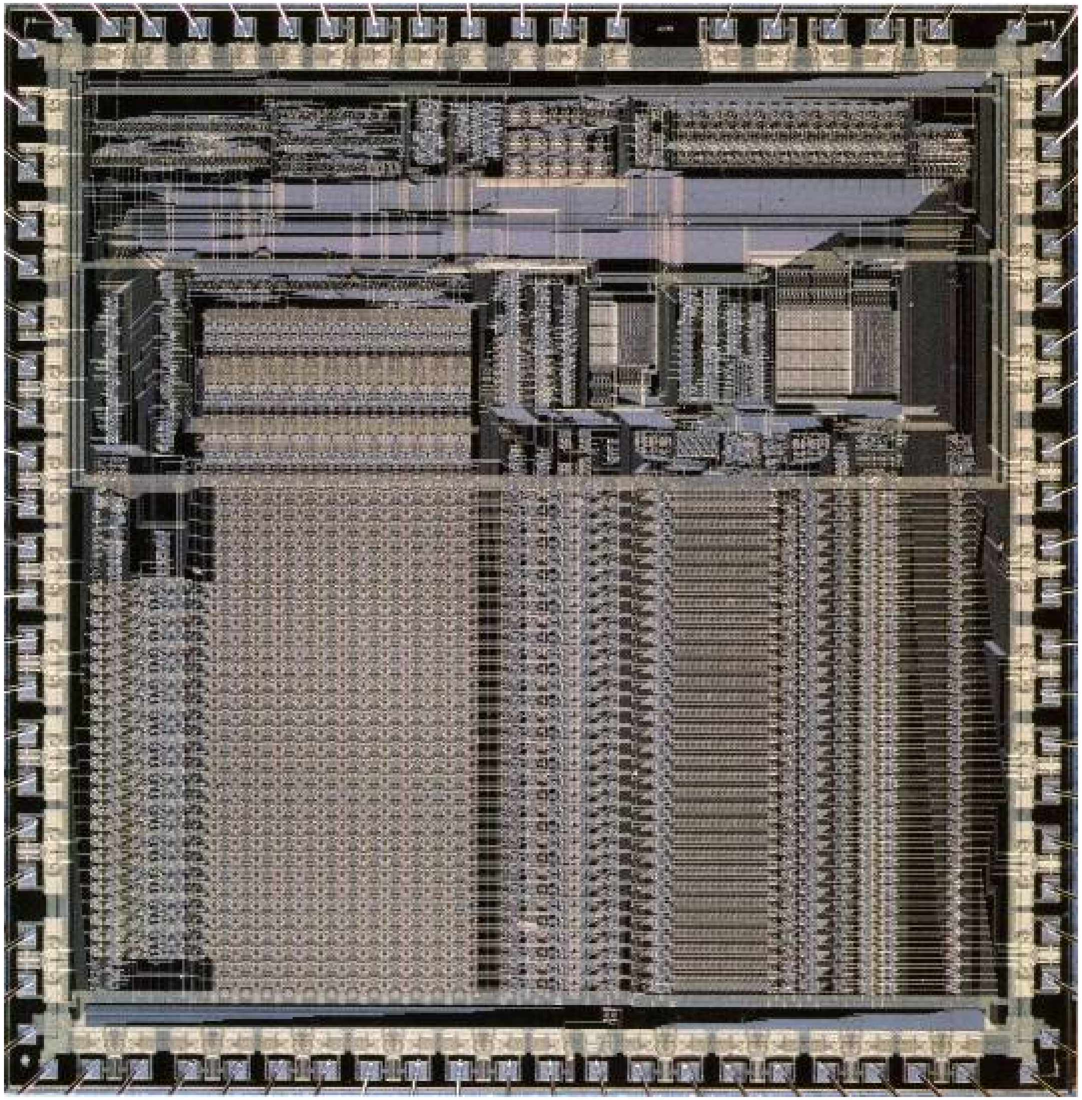


1906

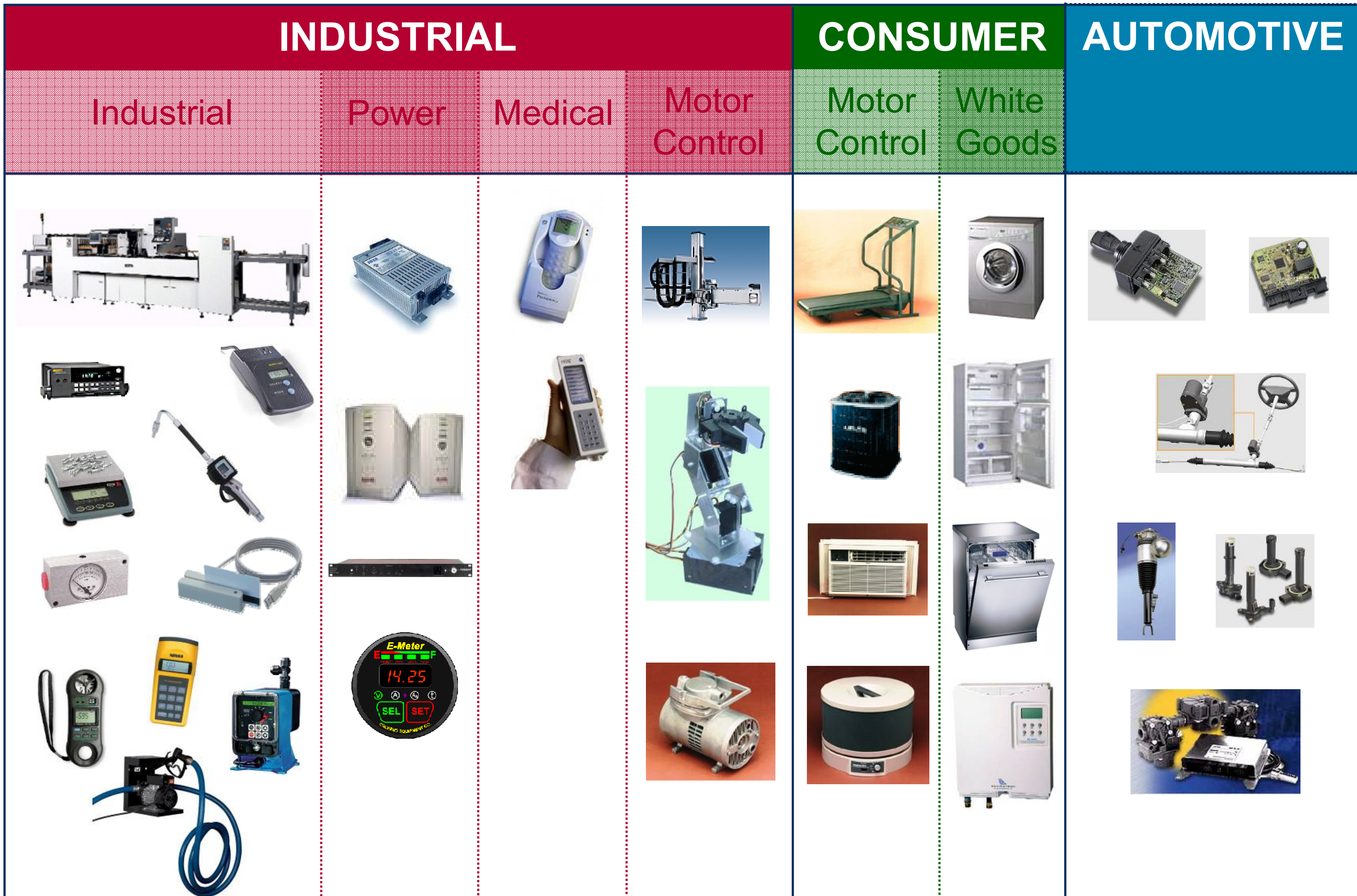
- San Francisco

2006

- ARM comes of age
  21 years old
- First ARM1 Silicon
  26th April 1985
- System architecture driven by packaging costs and no fan
- Cortex™-M3 returns to simplicity of the original design

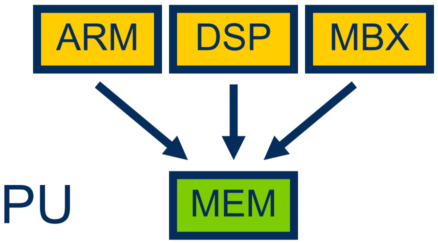


Vast Base of Embedded Applications

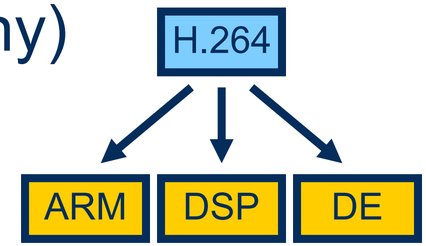


System Architecture Problems Today

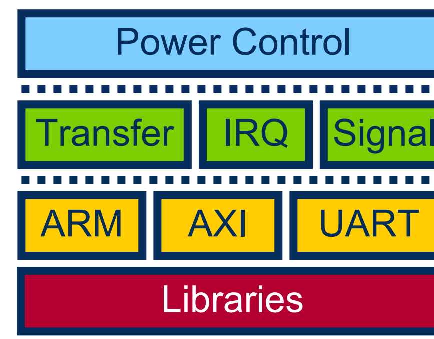
- Resource Sharing (horizontal, many to one)
  - Multiple IP blocks come together on a SoC
  - IP blocks are shared e.g. Memory, busses, DSP, CPU
  - Protection, security, QoS



- Application Partitioning (horizontal, one to many)
  - Applications require distributed resources
  - Core + DSP, multi-processing, accelerators
  - Signalling, communication, single-source compiler



- System Layering (vertical, many to many)
  - Enable differentiation and “Technology Towers”
  - Portability, compatibility, reliability
  - Start-up configuration, Discovery, PM



Leverage an Embedded Ecosystem

Over 30,000 members on ARM-based MCUs

Silicon available through online distributors

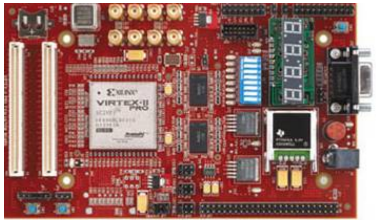
Major franchise distributors have teams of FAEs familiar with the ARM architecture

Quality as well as Quantity: Many of these third parties identify ARM related business as ‘largest growth driver’, which means robust, supported solutions

TOOLCHAINS PLATFORMS	DEBUGGERS	OPERATING SYSTEMS

Significance of FPGA increasing

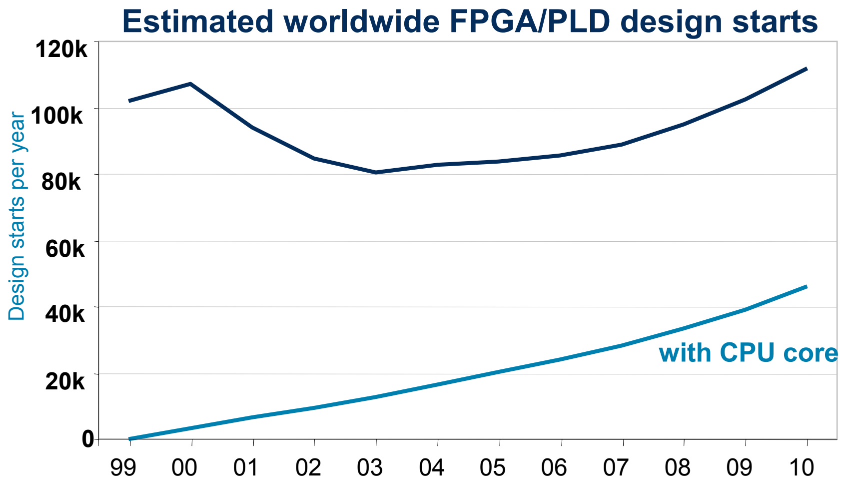
- FPGA market trends
  - Decreasing costs - 25% year on year*
  - Increasing capacity – 90nm, 65nm…
  - 40k CPU-based design starts in 2009**



- ARM7TDMI® processor on FPGA via Actel
  - 2005 Product of the Year from Electronic Products; EDN Hot 100

"The market will experience faster growth if an increasing number of large OEMs accept embedded processor cores."

Gartner

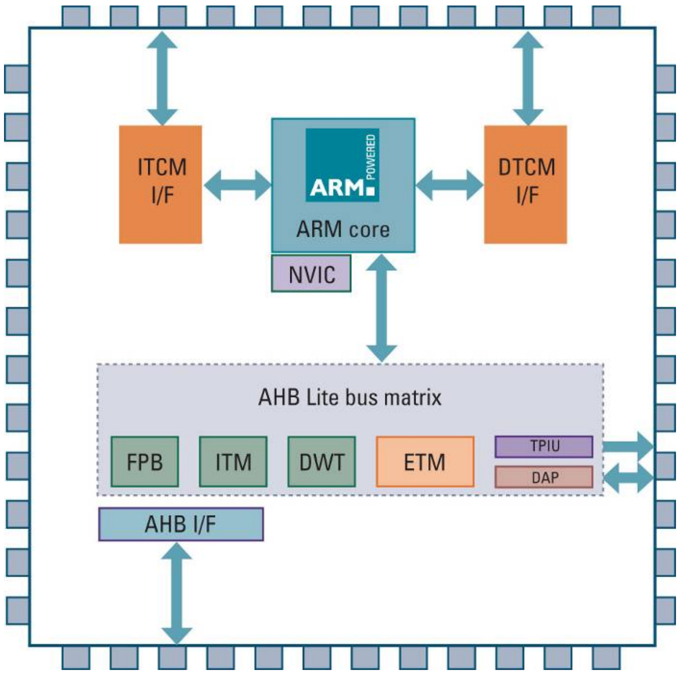


* Source: Altera
** Source: Xilinx

Proteus for FPGA

- An ARM CPU specifically optimised for FPGA
  - ARM architecture, quality and ecosystem on FPGA
  - FPGA flexibility, time-to-market and economics

- Release to lead Partners Q4’06



- Proteus specification
  - Classic Thumb® ISA
  - 3-stage Harvard architecture
  - Targets all popular FPGA devices
  - Upward compatible with Cortex™-M3
  - Integrated interrupt controller
  - Optional debug and trace

- Demo in the ARM Powered house

The Battery is Never Your Friend

- My first Panasonic 1991
  - 24h stand by 430mAh using NiCad

64K ROM + 16K RAM

- My first ARM7TDMI Nokia 6110
  - 270h stand by 900mAh NiMH

- My many much loved Nokia 6310i
  - 400h stand by 1100mAh Lithium-ion




- My first Smart Phone
  - 140h stand by 1050mAh Lithium-ion

32M ROM + 64M RAM

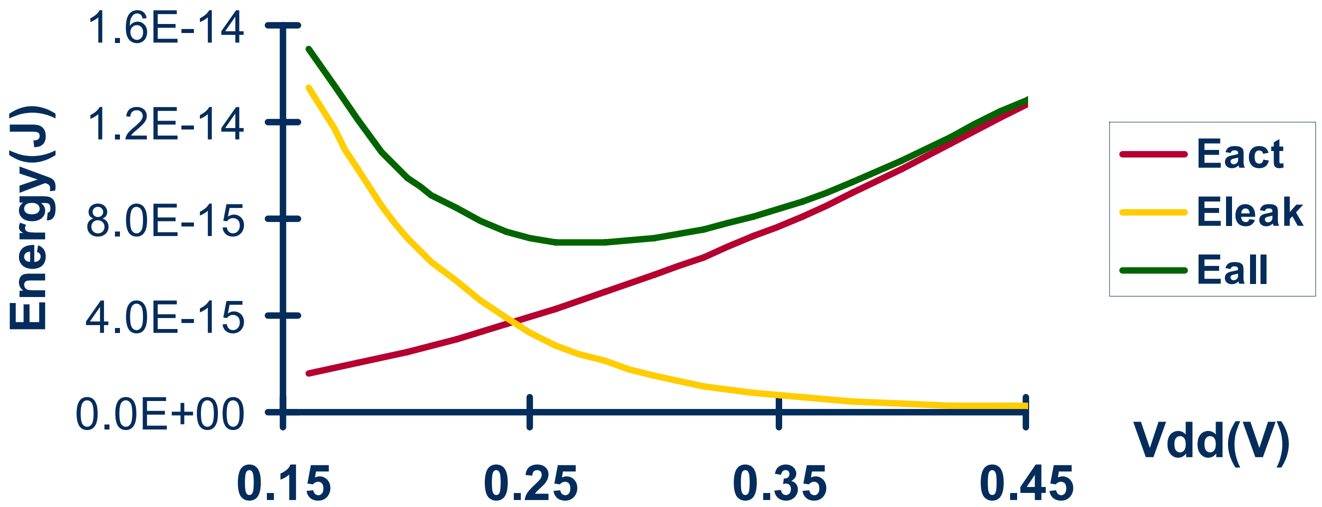
- My current phone
  - 250h stand by 1150mAh Lithium-ion

64M ROM + 64M RAM + 2GB SD

- 10 phones in 15 year = 1 every 18 months……..
- But I'm worried about my son

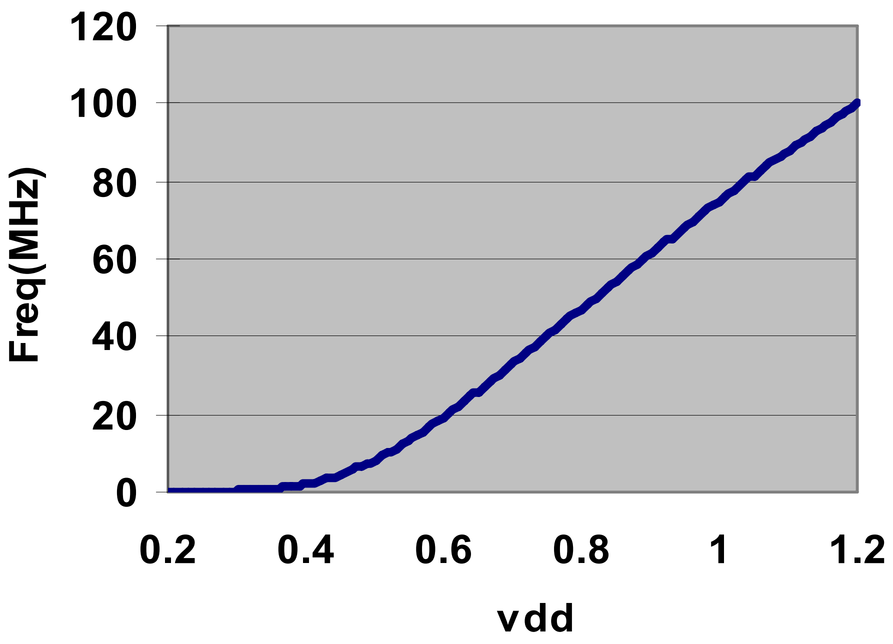
If You Really Want Low Power…..

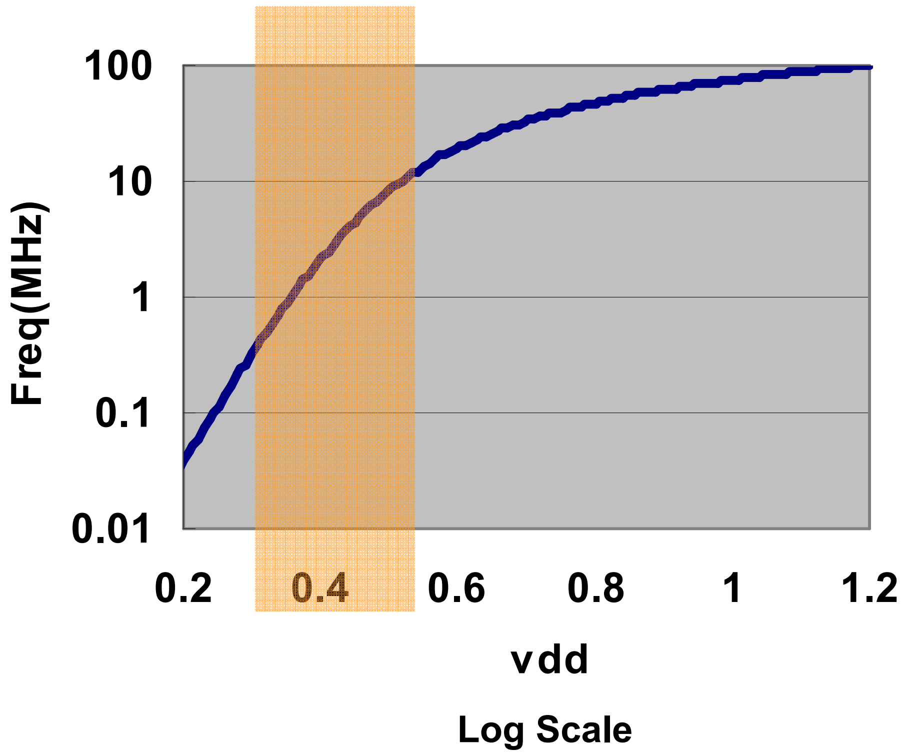
- Super-threshold region
  - Delay scales linearly and active power scales cubically
  - Leakage power changes little
  - Active power dominates leakage power
- Sub-threshold region
  - Delay and leakage energy scale exponentially
  - Active energy becomes comparable with leakage energy



But You Still Want Performance

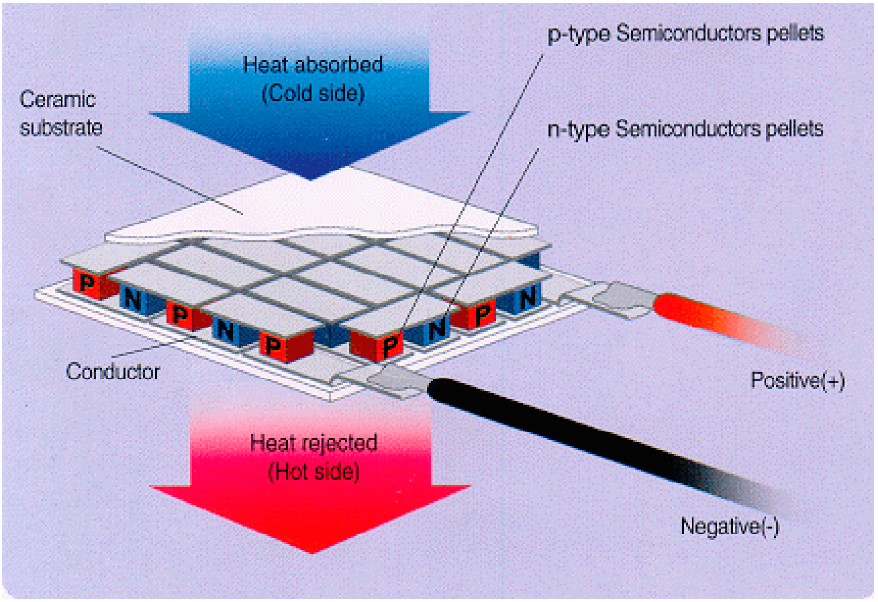
- Very dependent on design techniques
  - Leakage control
  - Performance of Logic and RAMs





Now You Need a Power Source

- Each thermocouple generate a few microvolts for each 1°C of temperature difference. Biophan plans to produce an array 2.5 centimetres square that generates 4 volts and delivers a power of 100 microwatts.



What's the Link?